

Catching Bitcoin Fraud with a Graph Convolutional Network

Using a GCN on the Elliptic dataset to spot illicit transactions

[Your name]

[roll no.]

Summary

Criminals use Bitcoin to move money from scams, ransomware and illegal online markets. Every Bitcoin transaction is public, so in principle we can follow the money, but there are far too many transactions to check by hand. In this project we train a **Graph Convolutional Network (GCN)** to look at a transaction *and the transactions connected to it* and decide whether it is **licit** (normal) or **illicit** (linked to crime).

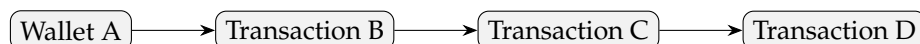
We use the real **Elliptic dataset** of about 200,000 Bitcoin transactions. We wrote the GCN ourselves in PyTorch instead of using a ready-made library layer, and we compared it with simpler models.

- Our GCN catches a little under half of the criminal transactions in the test period, and when it raises an alarm it is right about 8 times out of 10.
- Using the connections between transactions makes the model's alarms much more trustworthy than the same model without them.
- A simple **Random Forest** model still did best overall, which matches what the creators of the dataset found.
- After a large illegal market was shut down, **every** model stopped working, because criminals changed their behaviour.

1 The Problem

Imagine you work at a bank or a crypto exchange. Thousands of Bitcoin transactions pass through every day, and a few of them carry money from crime. You need to flag those few for an investigator to look at, without drowning the investigator in false alarms.

The key idea of this project is simple: **a transaction is known by the company it keeps**. Money does not appear out of nowhere. It comes from earlier transactions and flows into later ones:



If B receives money from known criminal transactions, B is suspicious too, even if B looks normal on its own. A normal machine-learning model looks at each transaction alone. A GCN also looks at its neighbours. Our plan was:

Problem: spot suspicious transactions → **Graph:** transactions and the money flowing between them →
GCN: learns from each transaction and its neighbours → **Output:** licit or illicit

2 The Dataset

The **Elliptic dataset** [2] was released by Elliptic, a company that tracks crypto crime, together with researchers from MIT and IBM. It is the largest public dataset of labelled Bitcoin transactions. We treat it as a graph:

- **Nodes (dots):** 203,769 Bitcoin transactions.
- **Edges (lines):** 234,355 money flows. An edge from A to B means money that came out of A was spent in B.
- **Features:** each transaction has 165 numbers describing it, such as how many inputs and outputs it has, the fee, and the amount. 93 of these describe the transaction itself, and the other 72 summarise its direct neighbours. The names of the features are hidden for privacy.
- **Time:** every transaction belongs to one of 49 time steps, each about two weeks apart.
- **Labels:** 4,545 are marked **illicit**, 42,019 are marked **licit**, and 157,205 (77%) are **unknown**. Nobody has checked those.

Figure 1 shows a real piece of the graph. About twenty illicit transactions all send money into one transaction. This “many-into-one” shape is a common way to collect criminal money. You can only see it by looking at the connections.

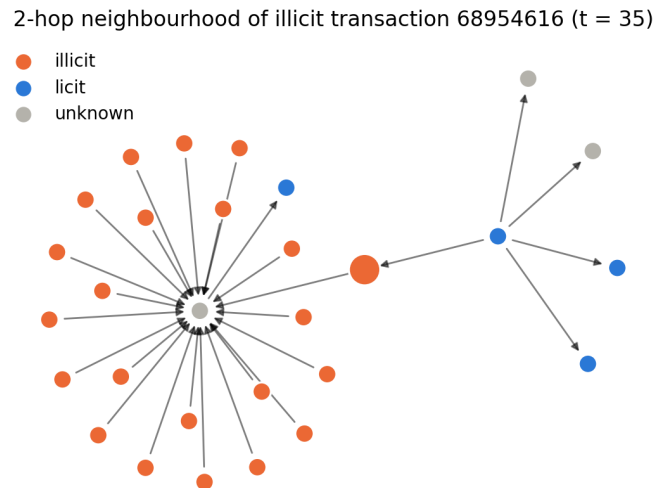


Figure 1: A real neighbourhood from the dataset. The big orange dot is an illicit transaction. Arrows show which way the money moved.

Three things that make this dataset tricky

1. **Very few criminals.** Only about 1 in 10 labelled transactions is illicit. A lazy model that says “everything is fine” would be 93.5% accurate on our test data while catching zero criminals. So **accuracy is a misleading score here**, and we use better ones (Section 4.4).
2. **Time matters.** We must predict the *future*, not the past. So we train on time steps 1–34 and test on time steps 35–49, as the dataset’s authors did. Transactions from different time steps are never connected, so the test transactions are in completely new graphs that the model has never seen.

3. **Neighbours really do look alike.** When we checked, a neighbour of an illicit transaction was illicit 54% of the time, compared with only 10% for a random labelled transaction. This tells us the connections carry useful information, which is exactly what a GCN needs.

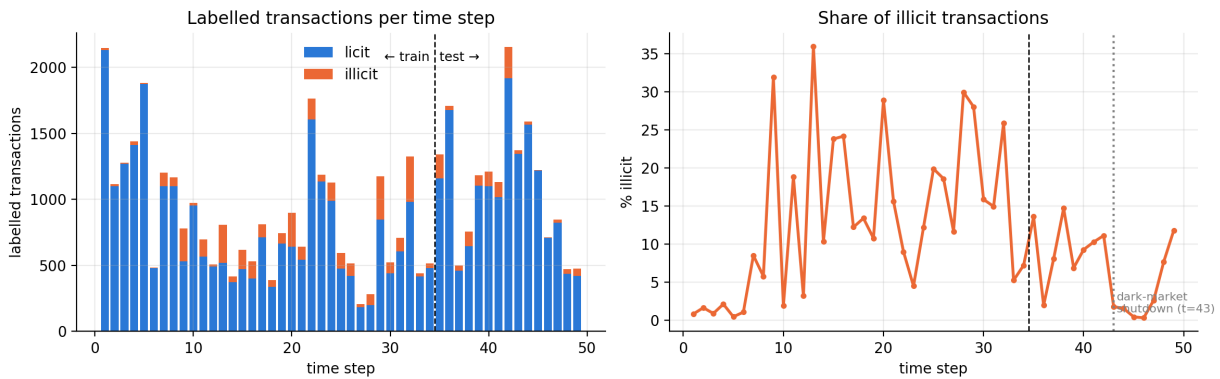


Figure 2: Left: labelled transactions in each time step. Right: the share that is illicit. The dashed line separates training (left) from testing (right).

Transactions (nodes)	203,769
Money flows (edges)	234,355
Features per transaction	165
Illicit / licit / unknown	4,545 / 42,019 / 157,205
Training data (time steps 1–34)	29,894 labelled, 3,462 of them illicit
Test data (time steps 35–49)	16,670 labelled, 1,083 of them illicit
Average number of neighbours	2.3

Table 1: The dataset in numbers.

3 What Is a GCN?

3.1 The idea

A normal neural network takes a list of numbers, such as one transaction’s features, and gives an answer. It cannot use the fact that transactions are connected.

A **Graph Convolutional Network** adds one simple step: **before deciding, every node looks at its neighbours and mixes their information with its own.** One round of mixing lets a node learn from its direct neighbours. Two rounds let it learn from its neighbours’ neighbours, two steps away. We use two rounds, called two *layers*.

In plain words: A GCN is like judging a person by their own behaviour *and* by the people they spend time with. One layer looks at friends; two layers also look at friends of friends.

3.2 Why use a GCN for this problem?

- **Fraud leaves a trail.** Criminal money is split, merged and passed along. Where the money came from and where it goes matters as much as the transaction itself.
- **Most transactions have no label.** Unlabelled transactions still pass information to their neighbours, so the model can use all 200,000 transactions, not just the labelled ones.

- **It is fast.** The model only does work along existing connections, so it handles 200,000 transactions in under a second on a normal laptop.

3.3 The formula, explained step by step

We describe the whole graph with an **adjacency matrix** A , a big table where $A_{ij} = 1$ if transactions i and j are connected and 0 otherwise. The features of all transactions form a table X with one row per transaction.

Step 1: Add self-loops. We connect every transaction to itself, $\tilde{A} = A + I$, so that when a node mixes in its neighbours' information, it also keeps its own.

Step 2: Make it fair. Some transactions have hundreds of neighbours and most have only two. If we simply added up neighbour information, busy nodes would get huge numbers. So we scale each connection by the number of neighbours on both sides:

$$\hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$$

which means that for two connected transactions i and j ,

$$\hat{A}_{ij} = \frac{1}{\sqrt{(\text{neighbours of } i + 1) \times (\text{neighbours of } j + 1)}}.$$

Here $\tilde{D}$ is a table that just stores how many connections each node has.

Step 3: One GCN layer. A layer does two things:

$$H_{\text{new}} = \text{ReLU}(\hat{A} \cdot H \cdot W)$$

- $H \cdot W$: transform each transaction's features using learned weights W . This is the part the model learns.
- $\hat{A} \cdot (\dots)$: replace each transaction by a weighted average of itself and its neighbours. This is the "look at the neighbours" part.
- ReLU: a simple function that sets negative numbers to zero, which lets the network learn non-linear patterns.

Step 4: Stack two layers and give an answer.

$$\text{Output} = \text{softmax}(\hat{A} \cdot \text{ReLU}(\hat{A} \cdot X \cdot W_1) \cdot W_2)$$

The softmax turns the final two numbers into two probabilities: "licit" and "illicit". If the illicit probability is at least 50%, we flag the transaction.

In plain words: Each layer is "average with your neighbours, then transform". The only things the computer learns are the two weight tables W_1 and W_2 . The graph part, $\hat{A}$, is fixed and calculated once from the connections.

3.4 Where does the formula come from? (short version)

GCNs come from *graph signal processing*. Just as a sound can be split into low and high frequencies, values on a graph can be split into "smooth" patterns, where neighbours have similar values, and "noisy" patterns, where neighbours differ. The exact version of this needs very heavy maths that is far too slow for 200,000 nodes. Kipf and Welling [1] showed that if you keep only the simplest possible approximation and add the self-loop trick from Step 1, you get exactly the "average with your neighbours" rule above. In other words, a GCN layer is a filter that keeps the smooth patterns and removes some of the noise.

4 How We Built It

4.1 Training the model

1. **Prepare the data.** Load the three files (features, labels, connections), scale every feature to a similar range, and build $\hat{A}$ from the list of connections.
2. **Make a guess.** Run all 203,769 transactions through the GCN to get an “illicit” probability for each one.
3. **Measure the mistakes.** Compare the guesses with the true labels, but only for the 29,894 labelled training transactions.
4. **Improve.** Adjust the weights slightly to reduce the mistakes, using the Adam optimiser.
5. **Repeat** steps 2–4 for 200 rounds (epochs).
6. **Test** once on time steps 35–49, which the model never saw during training.

Dealing with so few criminals. Because only about 1 in 10 examples is illicit, the model could get a low error by ignoring them. To stop this, we count a mistake on an illicit transaction as more serious than a mistake on a licit one (weights 0.7 and 0.3, as in [2]). This is called a *weighted loss*.



Figure 3: How data moves through our two-layer GCN.

4.2 The code

We wrote the GCN ourselves in PyTorch, so every line matches the formula. The two key pieces are shown here; the full model code is in the Appendix (Section 9). Building $\hat{A}$:

```
def build_norm_adj(src, dst, n):
    loops = torch.arange(n)
    row = torch.cat([src, dst, loops]) # connections in both directions + self-loops
    col = torch.cat([dst, src, loops])
    key = torch.unique(row * n + col) # remove duplicates
    row, col = key // n, key % n
    d = torch.zeros(n).scatter_add_(0, row, torch.ones(row.numel())) # count neighbours
    w = d[row].pow(-0.5) * d[col].pow(-0.5) # fair scaling
    return torch.sparse_coo_tensor(torch.stack([row, col]), w, (n, n)).coalesce()
```

One GCN layer is a single line, “average the transformed features over the neighbours”. This is slightly simplified from the notebook, which also reuses a pre-computed $\hat{A}X$ to save time:

```
class GCNLayer(nn.Module):
    def __init__(self, in_dim, out_dim):
        super().__init__()
        self.W = nn.Parameter(torch.empty(in_dim, out_dim))
        self.b = nn.Parameter(torch.zeros(out_dim))
        nn.init.xavier_uniform_(self.W)
    def forward(self, A, H):
        return torch.sparse.mm(A, H @ self.W) + self.b # A_hat (H W) + b
```

A note on size. Storing $\hat{A}$ as a full table would need $203,769 \times 203,769$ numbers, about 166 GB of memory. But almost all of those numbers are zero, because each transaction has only a couple

of neighbours. So we store only the 672,479 non-zero entries as a *sparse* matrix, which fits easily in memory. The whole model has just 16,802 numbers to learn.

Size of the hidden layer	100
Optimiser	Adam, learning rate 0.01
Training rounds	200
Loss	weighted, 0.7 for illicit and 0.3 for licit
Flag a transaction if	illicit probability $\geq 50\%$
Repeated with	5 different random starting points (seeds)

Table 2: Settings. We chose these in advance and did not tune them on the test data.

4.3 What we compared against

To check whether the graph really helps, we also trained:

- **MLP:** exactly the same network but *without* the neighbour step ($\hat{A}$ replaced by “do nothing”). This is the fairest comparison, because the only difference is the graph.
- **Skip-GCN:** a GCN with an extra shortcut, so that each transaction’s own features also go straight to the output and are not diluted by its neighbours.
- **Logistic Regression** and **Random Forest:** two classic models that look at each transaction alone.

We ran every model twice: once with all 165 features, and once with only the 93 features that describe the transaction itself.

4.4 How we score the models

Since accuracy is misleading here, we use the scores a fraud team would actually care about:

- **Precision:** of the transactions we flagged, how many were really illicit? Low precision means wasting investigators’ time on false alarms.
- **Recall:** of all the illicit transactions, how many did we catch? Low recall means criminals slipping through.
- **F1 score:** one number that balances precision and recall. It is high only if both are high. This is our main score.
- **Ranking score (AP):** if we sort all transactions from “most suspicious” to “least suspicious”, how well are the illicit ones pushed to the top? This does not depend on the 50% cut-off.

5 Results

5.1 Training

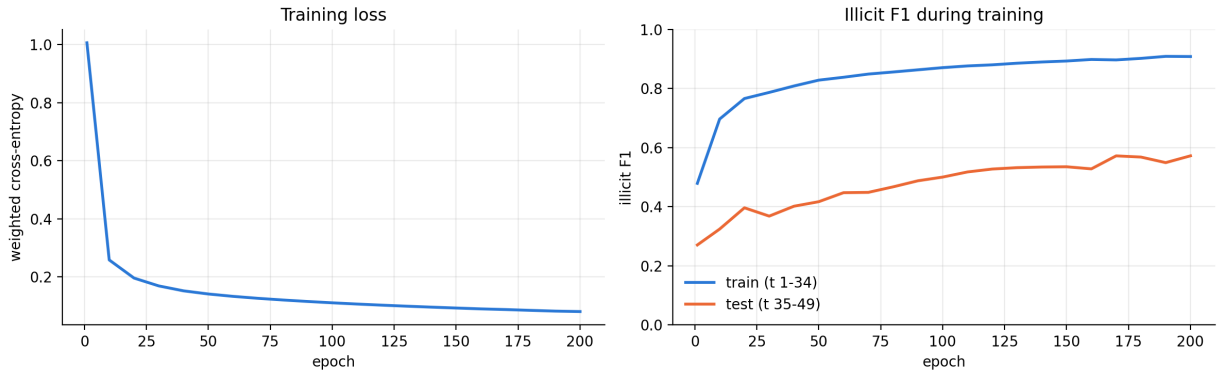


Figure 4: Left: the error goes down as the model learns. Right: F1 score on the training data (blue) and on the unseen test data (orange).

The error drops quickly in the first 20 rounds and then keeps falling slowly (Figure 4). The F1 score on the training data reaches 0.91, while on the unseen test data it climbs to 0.57. The gap is normal: the future is always harder to predict than the past.

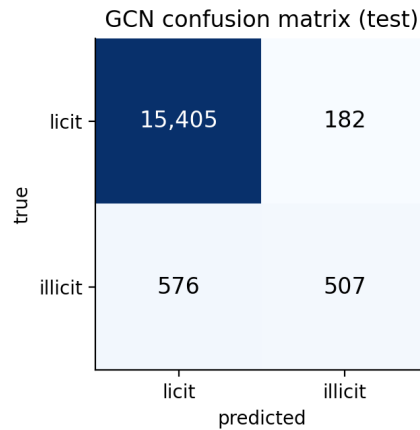


Figure 5: What the GCN got right and wrong on the 16,670 test transactions.

Figure 5 shows the GCN's decisions on the test data. It flagged 689 transactions, and 507 of them really were illicit, so **74% of its alarms were correct**. It caught 507 of the 1,083 illicit transactions (47%) and missed the rest.

5.2 Comparing all the models

Features used	Model	Precision	Recall	F1 score	Ranking (AP)
All 165	Logistic Regression	0.31	0.68	0.43	0.32
	MLP (no graph)	0.56	0.54	0.55	0.44
	GCN (ours)	0.79	0.44	0.56	0.60
	Skip-GCN	0.76	0.47	0.58	0.60
	Random Forest	0.92	0.72	0.81	0.78
Own 93 only	Logistic Regression	0.39	0.71	0.51	0.30
	MLP (no graph)	0.45	0.75	0.56	0.52
	GCN (ours)	0.43	0.60	0.50	0.53
	Skip-GCN	0.39	0.62	0.48	0.49
	Random Forest	0.85	0.71	0.77	0.77

Table 3: Test results, averaged over 5 runs (higher is better for every column).

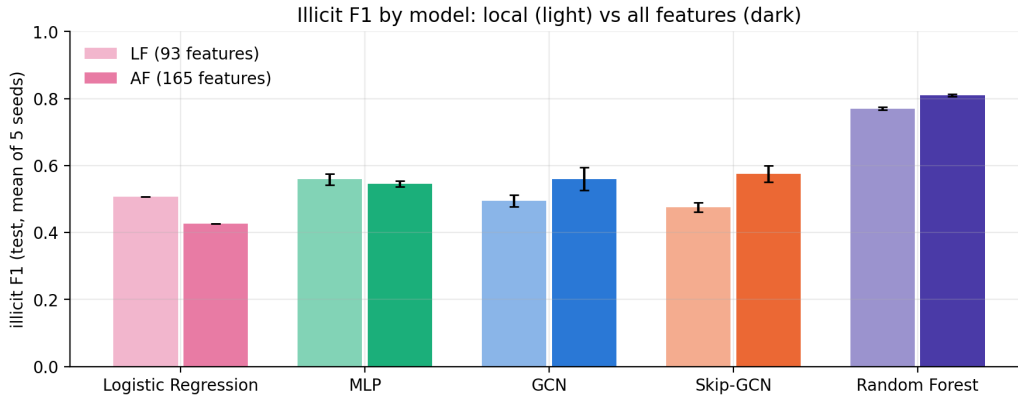


Figure 6: F1 score of each model. Light bars use only a transaction’s own 93 features; dark bars use all 165.

5.3 Results over time

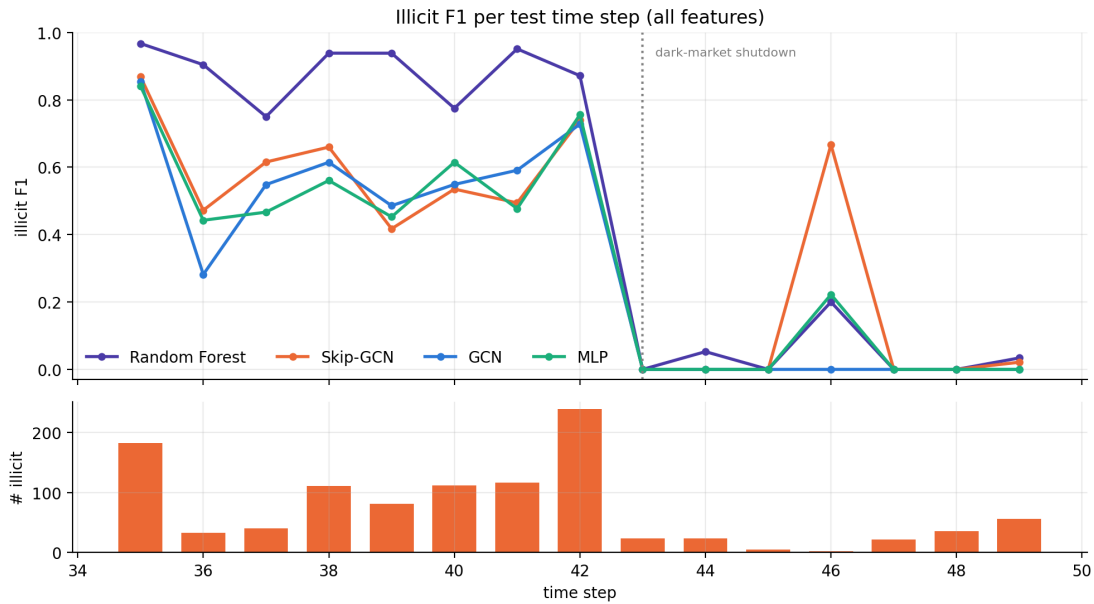


Figure 7: F1 score of each model in each test time step (top), and how many illicit transactions each time step had (bottom).

5.4 Does adding more layers help?

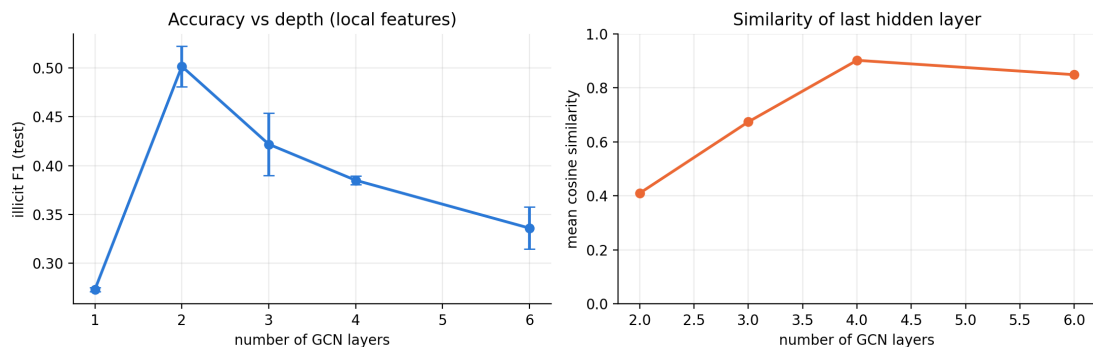


Figure 8: Left: F1 score as we add more GCN layers. Right: how similar the transactions look to each other inside the model (1 = all identical).

6 What We Learned

6.1 The graph makes alarms more trustworthy

The fairest test of the graph is GCN versus MLP, because they are the same network and the only difference is whether it looks at neighbours. With all features:

- **Precision rose from 56% to 79%**, so far fewer false alarms.
- **The ranking score rose from 0.44 to 0.60**, so illicit transactions are pushed much closer to the top of the list.
- **But recall dropped from 54% to 44%**, so the F1 score only improved a little (0.55 to 0.56).

In other words, looking at neighbours made the model more careful and more accurate when it raised an alarm, but also a bit more hesitant to raise one. In practice, a bank could lower the 50% cut-off to catch more criminals while keeping fairly good precision.

6.2 But the graph is not magic

When we gave the models only each transaction's own 93 features, the GCN did *not* beat the MLP on F1 (0.50 versus 0.56). The original researchers saw the same thing [2]. We think there are three reasons:

- **Too much averaging.** Most transactions have only about two neighbours, so when a node averages with them, half or more of its own information is replaced by theirs. Since 77% of transactions are unlabelled, much of what gets mixed in carries no clear signal.
- **Criminal transactions often touch normal ones.** Only about half of an illicit transaction's labelled neighbours are illicit, so averaging can water down exactly the rare signal we want to find.
- **The future looks different.** The test graphs are new, and the connection patterns learned from the past carry over less well than simple per-transaction facts.

6.3 The simple model won

The Random Forest scored best (F1 0.81), just as in the original study. Random Forests are very good at tables of mixed numbers such as amounts, fees and counts, and they can learn sharp rules like “fee above X and more than Y outputs”. This is a useful reminder that a fancier model is not always better. The best choice depends on the data.

6.4 Every model failed when criminals changed their behaviour

Figure 7 may be the most important result. Up to time step 42, all models work reasonably well. At time step 43 a big illegal online market was shut down, and from that point **every model's F1 score falls to almost zero**. (The jump at time step 46 is just luck: that time step had only 2 illicit transactions.)

Criminals simply started doing things differently, and models trained on old patterns could not recognise the new ones. This is called **concept drift**. It means a real fraud-detection system must be retrained regularly with fresh data. No model choice can fix this on its own.

6.5 Two layers is the sweet spot

Figure 8 shows what happens with more layers. One layer is too simple (F1 0.27), two layers work best (0.50), and after that the score keeps falling (0.42, 0.39, 0.34). At the same time, the transactions start to look more and more alike inside the model: their similarity rises from 0.41 to 0.90.

This is called **over-smoothing** [3]. Every layer averages each node with its neighbours. Do this many times and everything in the same area is averaged into the same grey mush, so illicit and licit transactions can no longer be told apart. That is why GCNs are usually kept to two or three layers.

In plain words: Averaging with your friends once gives you useful context. Averaging with friends of friends of friends makes everyone in town look the same.

7 Limitations

- **We did not tune the settings.** We used standard values to keep the test fair. With tuning, the GCN would probably score higher; the original paper reached 0.63 with longer training.
- **Only 5 runs.** Small differences of about 0.02 in F1 between the neural models could be down to chance.
- **We ignored the direction of money flow.** We treated “A pays B” the same as “B pays A”. Keeping direction could help.
- **All neighbours count the same.** A smarter model, such as a Graph Attention Network, could learn which neighbours matter more.
- **The model does not update itself over time,** so it cannot cope with changes in criminal behaviour (concept drift).

8 Conclusion

We built a Graph Convolutional Network from scratch and used it on a real problem: finding criminal Bitcoin transactions among more than 200,000. The GCN uses a simple idea, that a transaction should be judged together with the transactions it is connected to.

Looking at neighbours made the model’s alarms much more reliable (precision 79% versus 56% without the graph), and we saw first-hand why GCNs are kept shallow. We also found the limits of the approach. A well-tuned Random Forest still won, and every model broke down when criminals changed their behaviour after a market shutdown. The main lesson is that graphs are a powerful extra source of information for fraud detection, but in the real world keeping the model up to date matters as much as choosing it.

References

- [1] T. N. Kipf and M. Welling. Semi-Supervised Classification with Graph Convolutional Networks. *ICLR*, 2017.
- [2] M. Weber, G. Domeniconi, J. Chen, D. K. I. Weidele, C. Bellei, T. Robinson and C. E. Leiserson. Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics. *KDD Workshop on Anomaly Detection in Finance*, 2019.
- [3] Q. Li, Z. Han and X.-M. Wu. Deeper Insights into Graph Convolutional Networks for Semi-Supervised Learning. *AAAI*, 2018.

9 Appendix: Model Code

This appendix lists the code of the model exactly as it appears in the notebook `gcn_elliptic.ipynb` (cells that only draw figures or print statistics are left out). Everything is plain PyTorch; no pre-built graph layer is used.

A.1 Setup and imports

Libraries, random seed and plot colours.

```
import os, time, random
import numpy as np
import pandas as pd
import scipy.sparse as sp
from scipy.sparse.csgraph import connected_components
import torch
import torch.nn as nn
import torch.nn.functional as F
import matplotlib.pyplot as plt
import networkx as nx
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (precision_score, recall_score, f1_score, accuracy_score,
                             confusion_matrix, average_precision_score)

def set_seed(seed):
    random.seed(seed); np.random.seed(seed); torch.manual_seed(seed)

set_seed(42)
torch.set_num_threads(os.cpu_count())
os.makedirs("figures", exist_ok=True)

# colours used in every figure
C_LICIT, C_ILLICIT, C_UNKNOWN = "#2a78d6", "#eb6834", "#b4b2ab"
C_MODEL = {"GCN": "#2a78d6", "Skip-GCN": "#eb6834", "MLP": "#1baf7a", "Random Forest": "#4a3aa7",
           "Logistic Regression": "#e87ba4"}
plt.rcParams.update({"figure.dpi": 110, "axes.spines.top": False, "axes.spines.right": False,
                    "axes.grid": True, "grid.alpha": 0.25, "font.size": 10})

print("PyTorch", torch.__version__, "| NumPy", np.__version__, "| pandas", pd.__version__)
```

A.2 Loading the Elliptic dataset

Reads the three CSV files (features, labels, edges). If they are missing, PyTorch Geometric is used only to download them.

```
FILES = ["elliptic_txs_features.csv", "elliptic_txs_classes.csv", "elliptic_txs_edgelist.csv"]

def locate_raw_files(data_dir="data/elliptic_bitcoin_dataset"):
    if all(os.path.exists(os.path.join(data_dir, f)) for f in FILES):
        return data_dir
    from torch_geometric.datasets import EllipticBitcoinDataset # downloader only
    EllipticBitcoinDataset("data/pyg_elliptic")
    return "data/pyg_elliptic/raw"

raw_dir = locate_raw_files()
t0 = time.time()
feat_df = pd.read_csv(os.path.join(raw_dir, FILES[0]), header=None)
class_df = pd.read_csv(os.path.join(raw_dir, FILES[1]))
edge_df = pd.read_csv(os.path.join(raw_dir, FILES[2]))
```

```
print(f"loaded from {raw_dir} in {time.time()-t0:.1f}s")
print("features:", feat_df.shape, "| classes:", class_df.shape, "| edges:", edge_df.shape)
```

A.3 Features, labels and edges

Turns the CSV files into a feature matrix, a label vector (−1 unknown, 0 licit, 1 illicit) and the list of money flows.

```
tx_ids    = feat_df[0].values
time_step = feat_df[1].values.astype(int)
X_all     = feat_df.iloc[:, 2:].values.astype(np.float32)      # 165 features
LOCAL, AGG = slice(0, 93), slice(93, 165)                    # 93 local + 72 aggregated

N = len(tx_ids)
pos = pd.Series(np.arange(N), index=tx_ids)                   # txId -> row index

label_str = class_df.set_index("txId").loc[tx_ids, "class"].astype(str).values
y_np = np.full(N, -1, dtype=np.int64)                       # -1 unknown, 0 licit, 1 illicit
y_np[label_str == "2"] = 0
y_np[label_str == "1"] = 1

src = pos[edge_df["txId1"].values].values
dst = pos[edge_df["txId2"].values].values

n_ill, n_lic, n_unk = (y_np == 1).sum(), (y_np == 0).sum(), (y_np == -1).sum()
print(f"transactions (nodes): {N:,}")
print(f"directed edges      : {len(src):,}")
print(f"features per node    : {X_all.shape[1]} (93 local + 72 aggregated)")
print(f"time steps          : {time_step.min()} .. {time_step.max()}")
print(f"illicit / licit / unknown : {n_ill:,} / {n_lic:,} / {n_unk:,}")
print(f"illicit share of labelled : {n_ill/(n_ill+n_lic):.2%}")
```

A.4 Train/test split and feature scaling

Train on time steps 1–34, test on 35–49. Features are scaled with the training-period mean and standard deviation.

```
train_mask_np = (y_np >= 0) & (time_step <= 34)
test_mask_np  = (y_np >= 0) & (time_step >= 35)

mu = X_all[time_step <= 34].mean(0); sd = X_all[time_step <= 34].std(0) + 1e-6
X_std = (X_all - mu) / sd

X_af = torch.tensor(X_std)      # all 165 features
X_lf = torch.tensor(X_std[:, LOCAL]) # 93 local features only
y     = torch.tensor(y_np)
train_mask, test_mask = torch.tensor(train_mask_np), torch.tensor(test_mask_np)

for name, m in [("train (t 1-34)", train_mask_np), ("test (t 35-49)", test_mask_np)]:
    print(f"{name}: {m.sum():6,} labelled | illicit {(y_np[m]==1).sum():5,} ({(y_np[m]==1).mean():.1%})")
print(f"\nAccuracy of always predicting 'licit' on test: {(y_np[test_mask_np]==0).mean():.2%} (↪ illicit F1 = 0)")
```

A.5 Building the normalised adjacency matrix $\hat{A}$

Adds self-loops, makes the graph undirected, and applies $\hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$ as a sparse matrix. The identity version turns the GCN into the MLP baseline.

```
def build_norm_adj(src, dst, n):
    src, dst = torch.as_tensor(src), torch.as_tensor(dst)
    loops = torch.arange(n)
    row = torch.cat([src, dst, loops])          # both directions + self-loops -> A~ = A + I
    col = torch.cat([dst, src, loops])
    key = torch.unique(row * n + col)           # remove duplicate edges
    row, col = key // n, key % n
    d = torch.zeros(n).scatter_add_(0, row, torch.ones(row.numel())) # D~_ii
    w = d[row].pow(-0.5) * d[col].pow(-0.5)     # D~^-1/2 A~ D~^-1/2
    return torch.sparse_coo_tensor(torch.stack([row, col]), w, (n, n)).coalesce()

def identity_adj(n):
    i = torch.arange(n)
    return torch.sparse_coo_tensor(torch.stack([i, i]), torch.ones(n), (n, n)).coalesce()

A_hat = build_norm_adj(src, dst, N)
I_hat = identity_adj(N)
print("A_hat:", tuple(A_hat.shape), "| non-zeros:", f"{A_hat._nnz():,}",
      f"| density {A_hat._nnz()/N**2:.2e}")
print("dense float32 size would be", f"{N*N*4/1e9:.0f} GB")

# sanity check: A_hat is symmetric and its largest eigenvalue is 1
v = torch.sparse.sum(A_hat, 1).to_dense()
print("row sums: min %.3f max %.3f" % (v.min(), v.max()))
```

A.6 The GCN model

Each layer computes $\hat{A}(HW) + b$. The same class builds the plain GCN, the Skip-GCN and deeper GCNs.

```
class GCNLayer(nn.Module):
    '''H_out = A_hat @ (H W) + b'''
    def __init__(self, in_dim, out_dim):
        super().__init__()
        self.W = nn.Parameter(torch.empty(in_dim, out_dim))
        self.b = nn.Parameter(torch.zeros(out_dim))
        nn.init.xavier_uniform_(self.W)
    def forward(self, A, H, AH=None):
        if AH is not None:
            # A_hat @ H already known: A_hat (H W) = (A_hat
            # -> H) W
            return AH @ self.W + self.b
        return torch.sparse.mm(A, H @ self.W) + self.b

class GCN(nn.Module):
    def __init__(self, in_dim, hid_dim, out_dim=2, n_layers=2, dropout=0.0, skip=False):
        super().__init__()
        dims = [in_dim] + [hid_dim] * (n_layers - 1) + [out_dim]
        self.layers = nn.ModuleList(GCNLayer(dims[k], dims[k + 1]) for k in range(n_layers))
        self.skip = nn.Linear(in_dim, out_dim, bias=False) if skip else None
        self.dropout = dropout
    def forward(self, A, X, AX=None, return_hidden=False):
        H, hidden = X, None
        for k, layer in enumerate(self.layers):
            if k == 0 and AX is not None and self.dropout == 0:
                H = layer(A, X, AH=AX) # first layer re-uses the pre-computed A_hat X
            else:
                H = layer(A, self.dropout(H, self.dropout, self.training))
```

```

        if k < len(self.layers) - 1:
            H = F.relu(H); hidden = H
        if self.skip is not None:
            H = H + self.skip(X)
        return (H, hidden) if return_hidden else H

m = GCN(165, 100)
print(m)
print("trainable parameters:", sum(p.numel() for p in m.parameters()))

```

A.7 Loss, training loop and metrics

Weighted cross-entropy (0.3 licit / 0.7 illicit), Adam optimiser, and precision / recall / F1 / AP on the test period.

```

CLASS_W = torch.tensor([0.3, 0.7])
HP = dict(hid=100, lr=1e-2, epochs=200, wd=5e-4, dropout=0.0)

def scores(y_true, y_pred, p_illicit=None):
    out = dict(precision=precision_score(y_true, y_pred, zero_division=0),
               recall=recall_score(y_true, y_pred, zero_division=0),
               f1=f1_score(y_true, y_pred, zero_division=0),
               micro_f1=accuracy_score(y_true, y_pred))
    if p_illicit is not None:
        out["ap"] = average_precision_score(y_true, p_illicit)
    return out

def train_gcn(A, X, seed=42, n_layers=2, skip=False, log_every=0, record=False, **hp):
    hp = {**HP, **hp}
    set_seed(seed)
    model = GCN(X.shape[1], hp["hid"], 2, n_layers=n_layers, dropout=hp["dropout"], skip=skip)
    opt = torch.optim.Adam(model.parameters(), lr=hp["lr"], weight_decay=hp["wd"])
    AX = torch.sparse.mm(A, X)  # A_hat and X never change, so A_hat X is computed once
    hist = []
    ytr, yte = y[train_mask], y[test_mask]
    for ep in range(1, hp["epochs"] + 1):
        model.train()
        out = model(A, X, AX)
        loss = F.cross_entropy(out[train_mask], ytr, weight=CLASS_W)
        opt.zero_grad(); loss.backward(); opt.step()
        if record and (ep == 1 or ep % 10 == 0):
            model.eval()
            with torch.no_grad():
                pred = model(A, X, AX).argmax(1)
            hist.append(dict(epoch=ep, loss=loss.item(),
                            train_f1=f1_score(ytr, pred[train_mask]),
                            test_f1=f1_score(yte, pred[test_mask]),
                            train_acc=(pred[train_mask] == ytr).float().mean().item(),
                            test_acc=(pred[test_mask] == yte).float().mean().item()))
        if log_every and (ep == 1 or ep % log_every == 0):
            h = hist[-1]
            print(f"epoch {ep:4d} | loss {h['loss']:.4f} | train F1 {h['train_f1']:.3f} | "
                  f"test F1 {h['test_f1']:.3f} | test acc {h['test_acc']:.3f}")
    model.eval()
    with torch.no_grad():
        logits, hidden = model(A, X, AX, return_hidden=True)
        prob = logits.softmax(1)[ :, 1]
    return model, prob.numpy(), hidden, pd.DataFrame(hist)

def eval_probs(prob, mask=test_mask_np):
    return scores(y_np[mask], (prob[mask] >= 0.5).astype(int), prob[mask])

```

A.8 Training the GCN

Trains the 2-layer GCN for 200 epochs with seed 42 and prints the test scores.

```
t0 = time.time()
gcn, p_gcn, H_gcn, hist_gcn = train_gcn(A_hat, X_af, seed=42, record=True, log_every=100)
print(f"\ntraining time: {time.time()-t0:.0f}s")
print("initial loss:", round(hist_gcn.loss.iloc[0], 4), " (ln 2 = 0.6931 for a 50/50 guess)")
res_gcn = eval_probs(p_gcn)
print({k: round(v, 4) for k, v in res_gcn.items()})
```

A.9 Baselines and repeated runs

Trains Logistic Regression, Random Forest, MLP, GCN and Skip-GCN on all features and on local features, with 5 seeds each.

```
def run_sklearn(kind, X, seed=42):
    if kind == "Logistic Regression":
        clf = LogisticRegression(max_iter=2000)
    else:
        clf = RandomForestClassifier(n_estimators=50, max_features=50, random_state=seed, n_jobs
        ↪=-1)
    clf.fit(X[train_mask_np], y_np[train_mask_np])
    prob = np.zeros(N); prob[test_mask_np] = clf.predict_proba(X[test_mask_np])[:, 1]
    return clf, prob

SEEDS = [42, 1, 2, 3, 4]
FEATS = {"AF": (X_std, X_af), "LF": (X_std[:, LOCAL], X_lf)}
NEURAL = [("MLP", I_hat, False), ("GCN", A_hat, False), ("Skip-GCN", A_hat, True)]

rows, probs = [], {}
t0 = time.time()
for feats, (Xn, Xt) in FEATS.items():
    _, p = run_sklearn("Logistic Regression", Xn)
    probs[("Logistic Regression", feats)] = p
    rows.append(dict(model="Logistic Regression", features=feats, seed=0, **eval_probs(p)))
    for s in SEEDS:
        _, p = run_sklearn("Random Forest", Xn, seed=s)
        if s == 42: probs[("Random Forest", feats)] = p
        rows.append(dict(model="Random Forest", features=feats, seed=s, **eval_probs(p)))
        for kind, A, skip in NEURAL:
            if (kind, feats, s) == ("GCN", "AF", 42):
                p = p_gcn # already trained in Section 7
            else:
                _, p, _, _ = train_gcn(A, Xt, seed=s, skip=skip)
                if s == 42: probs[(kind, feats)] = p
                rows.append(dict(model=kind, features=feats, seed=s, **eval_probs(p)))
        print(f"{feats} seed {s:2d} done ({time.time()-t0:.0f}s)")

multi_df = pd.DataFrame(rows)
multi_df.to_csv("figures/all_runs.csv", index=False)
```